NovaCart Commerce

NovaCart serves customers through a unified digital platform, offering products ranging from electronics, fashion, home essentials, and personal care to groceries, fresh produce, beverages, and daily necessities. Customers can purchase products through standard e-commerce deliveries or rapid deliveries fulfilled by nearby dark stores and local fulfillment hubs.

Company Profile

Company type

Omnichannel digital commerce platform combining marketplace and quick-commerce operations.

Primary customers

Retail consumers, households, working professionals, families and time-sensitive shoppers.

Business ecosystem

Customers, marketplace sellers, inventory operators, dark-store teams, delivery partners, payment providers, customer support and finance.

Strategic priorities

Fast delivery, reliable fulfillment, customer trust, accurate payments, low operational cost and controlled refund leakage.

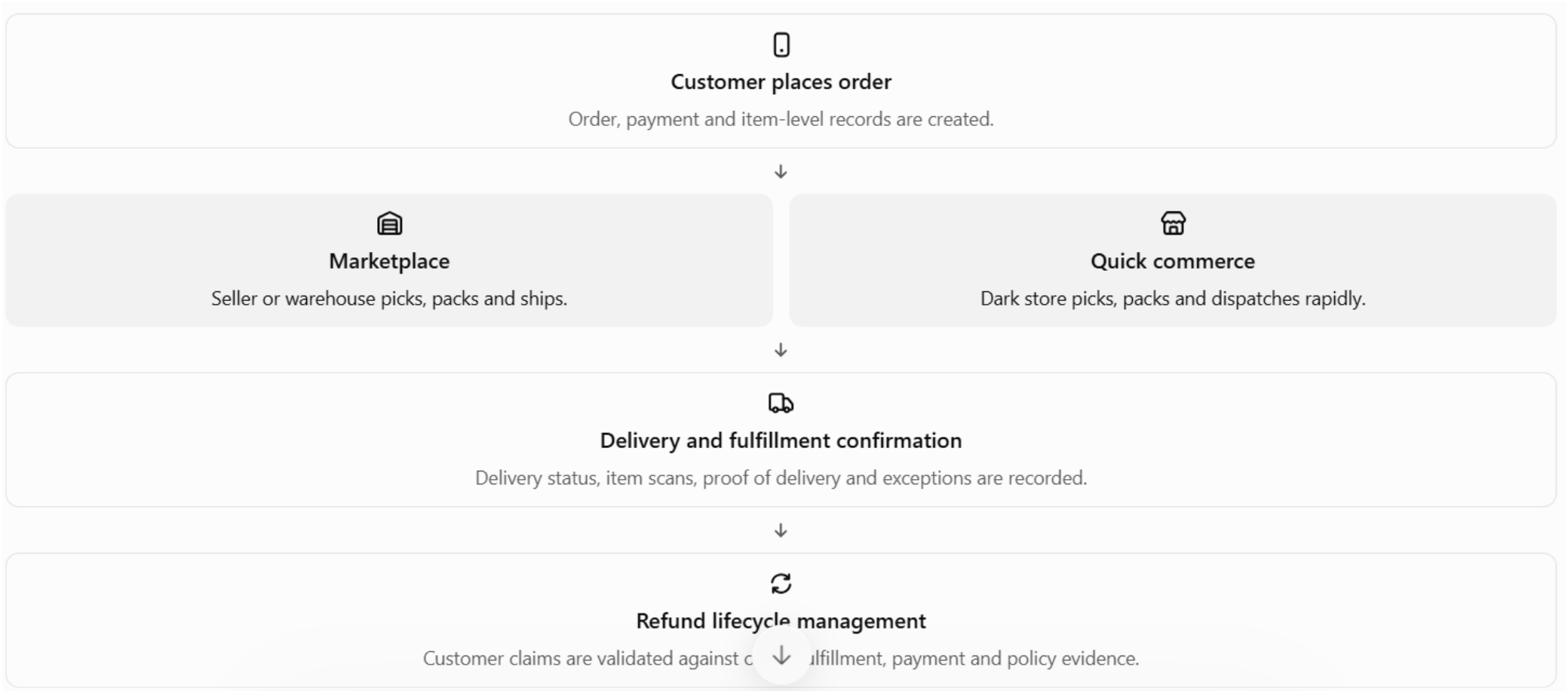
Company objected and Expected outcomes

Objective	Expected outcome
Automate refund eligibility	Reduce manual checks through policy-driven validation
Detect duplicate payments and claims	Prevent refunds exceeding the eligible amount
Apply AI-based fraud detection	Identify suspicious transactions for additional review
Accelerate refund processing	Reduce turnaround time for eligible refunds
Automate payment and reconciliation	Minimize manual intervention and unmatched transactions
Improve customer experience	Provide transparent, timely refund status updates
Establish auditability	Maintain a complete record of decisions and transactions

Problem Statement

- The proposed **AI-Powered Refund Management System** will act as a centralized intelligence and orchestration layer that manages the complete refund lifecycle—from customer initiation to eligibility assessment, risk evaluation, approval, payment execution, settlement reconciliation, and final closure.
- The objective is not simply to automate refund approvals. It is to create a controlled, auditable, and scalable system that balances customer satisfaction, financial accuracy, operational efficiency, and fraud prevention.
- As the business scales, refund management becomes a critical operational function. Refund requests can originate from damaged products, missing items, incorrect deliveries, cancellations, failed payments, partial fulfillment, quality issues, and returns. These requests must be validated against order history, payment records, fulfillment events, return policies, fraud indicators, and payment settlement data.

Fulfillment and delivery



Revenue and financial flows

- NovaCart may generate revenue through:
  - Product sales and applicable margins.
  - Marketplace commissions or seller service fees.
  - Delivery, convenience or service fees where applicable.
  - Promotional partnerships and other commerce services.
- A refund may affect several financial components, not just the product price. The refund engine must understand discounts, coupons, taxes, delivery charges, wallet credits, promotional benefits, seller settlements and any applicable adjustments.
- The exact allocation rules must be approved by Finance and Legal. They must not be inferred by the AI model.

Why refund management is a critical business function

Business scenario	Example	Refund-management challenge
Failed payment	Customer is debited but order confirmation fails	Determine whether funds were captured, reversed or settled
Order cancellation	Order cancelled before dispatch	Calculate the refundable amount and verify payment state
Missing item	One product is absent from a multi-item order	Calculate the correct item-level refund
Damaged product	Electronic device arrives damaged	Apply category-specific policy and evidence requirements
Incorrect item	A different SKU is delivered	Validate ordered versus delivered product
Quality issue	Fresh produce is spoiled	Use item, fulfillment and claim evidence
Return request	Customer sends back an eligible product	Track pickup, return receipt and inspection where required
Duplicate refund	Two support agents initiate refunds for the same item	Prevent multiple financial executions
Fraudulent claim	Repeated high-value claims show suspicious patterns	Assess risk without automatically treating a customer as fraudulent
Settlement mismatch	Refund API reports success but ledger records disagree	Reconcile payment-provider and internal financial records

Refund lifecycle: end-to-end business process

1. **Refund request initiation**

   Accept requests from the app, website, chatbot or customer support. Capture order ID, item, quantity, reason, requested resolution and evidence.

2. **Identity and transaction verification**

   Validate customer access, order existence, payment reference, delivery state and prior refund history.

3. **Policy and eligibility validation**

   Evaluate the applicable policy, category restrictions, request window, fulfillment status, seller responsibility and eligible amount.

4. **Duplicate claim and financial checks**

   Check previous claims, refund reservations, executed refunds, payment reversals and remaining refundable balance.

5. **AI-based fraud and anomaly assessment**

   Estimate risk from relevant behavioral, transactional and operational signals. Route uncertain or high-risk cases for review.

6. **Decision and approval**

   Apply deterministic rules and approval limits. Automatically approve eligible low-risk cases and escalate exceptions.

7. **Refund execution**

   Initiate a refund through the appropriate payment provider or internal wallet mechanism, using idempotency and transaction controls.

8. **Settlement and reconciliation**

   Compare refund instructions, provider responses, settlement reports and ledger entries. Resolve pending or unmatched transactions.

9. **Customer notification and closure**

   Communicate the current status, record the final outcome, preserve audit evidence and close only when required completion conditions are satisfied.

Business rules versus AI decisions

Capability	Primary mechanism	Expected responsibility
Policy eligibility	Rules engine	Determine whether the claim satisfies approved policy conditions
Refund amount	Deterministic calculation service	Calculate the maximum eligible amount
Duplicate prevention	Transaction controls and ledger checks	Prevent repeated or excessive refunds
Fraud assessment	ML risk model	Estimate the probability of suspicious activity
Evidence interpretation	AI-assisted classification	Extract relevant information from customer text, images or support notes
Approval routing	Workflow engine	Apply risk bands, financial thresholds and escalation rules
Payment execution	Payment orchestration service	Submit controlled refund instructions
Settlement reconciliation	Financial matching engine	Match internal records with payment-provider and bank records
Customer communication	Template-based messaging with AI assistance	Explain the verified status in clear language
Audit and compliance	Immutable or tamper-evident audit records	Preserve decision history and transaction traceability

AI should assist with interpretation, prioritization and anomaly detection. Approved business rules and transaction controls should remain authoritative for financial eligibility, amount calculation, duplicate prevention and payment execution.

7. Refund scenarios the system must support



Scenario A: Missing grocery item

Potential auto-approval

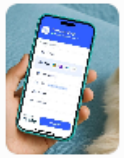
A customer reports a missing item in a completed quick-commerce order. The system checks the item-level order record, picking or packing evidence, substitutions, prior claims and applicable policy. If the evidence and rules support the claim, it can approve a partial refund.



Scenario B: Damaged electronics

Evidence-based review

A customer reports a damaged electronic product. The system validates purchase details, category-specific return rules, serial number where applicable, customer evidence and return requirements. High-value or ambiguous claims may require manual inspection or approval.



Scenario C: Payment failure or duplicate debit

The customer reports being charged twice or charged for an unsuccessful order. The system retrieves payment attempts, transaction identifiers, reversals and provider status. It refunds only the verified excess or eligible captured amount, avoiding a second refund where an automatic reversal has already occurred.



Scenario D: Return pickup and refund

A customer returns an eligible product. The system tracks pickup, receipt, inspection and any category-specific release conditions. The refund is released according to the approved policy and the applicable return state.



Scenario E: Refund initiated but not reconciled

The provider reports a refund operation, but the internal ledger or settlement file does not show a matching record. The case enters an exception queue, is investigated using transaction references and is not blindly retried.

Business stakeholders and ownership

The FDE must establish who owns each decision, data source and operational exception.

Stakeholder	Responsibility
Customer experience	Customer-facing refund journeys and communication
Refund operations	Manual reviews, exception queues and operational procedures
Finance and accounting	Refund accounting, ledger accuracy and settlement reconciliation
Payments team	Payment gateways, UPI, cards, wallets, reversals and provider integrations
Marketplace operations	Seller coordination, return policies and dispute resolution
Quick-commerce operations	Dark-store picking, item shortages, substitutions and delivery exceptions
Fraud and risk	Risk policies, investigation, model monitoring and fraud escalation
Product and engineering	APIs, workflow integration, service ownership and user experience
Data science and ML	Training data, model development, evaluation and monitoring
Legal, privacy and compliance	Consumer policy, data protection, audit and applicable regulatory obligations

Every workflow stage needs an explicit owner, service-level target, escalation path and failure-recovery procedure.

Business KPIs and expected outcomes

KPI	Definition	Illustrative initial target
Straight-through processing	Eligible refunds completed without manual intervention ÷ eligible requests	60–80%
Refund turnaround time	Time from request creation to the defined refund completion milestone	Reduce by 30–50%
Duplicate refund rate	Confirmed duplicate refunds ÷ completed refunds	Near zero
Refund accuracy	Refunds with correct policy, amount and transaction outcome	At least 99.5%
Reconciliation rate	Refund transactions reconciled within the agreed SLA ÷ transactions due for reconciliation	At least 99%
Fraud loss rate	Confirmed fraudulent refund losses relative to a defined refund or transaction denominator	Reduce against baseline
Manual review rate	Requests routed for human review ÷ total requests	Establish baseline; optimize safely
Customer contact rate	Refund-related support contacts per 1,000 orders	Reduce against baseline

Definition of success

NovaCart's refund management initiative will be successful when:

- Eligible refunds can be processed automatically within approved policy and risk limits.
- No refund can exceed the remaining eligible amount, and concurrent requests cannot cause duplicate financial execution.
- High-risk, ambiguous and policy-exception cases are routed to authorized reviewers with sufficient evidence.
- Payment execution and financial reconciliation remain independently verifiable.
- Customers receive accurate, timely status updates without unsupported promises about settlement dates.
- Finance and operations can trace every refund from the original order through its final financial outcome.
- AI decisions are monitored for accuracy, bias, drift and unintended customer impact.

The long-term goal is a **policy-controlled, AI-assisted refund platform that scales with both marketplace and quick-commerce operations without sacrificing financial integrity or customer trust**.